Topic 2 — Basic Concepts and Rules of Probability

Introduction to Probability Theory

Probability Theory is a branch of mathematics that deals with uncertainty and randomness. It provides a framework for quantifying and analyzing uncertainty in stochastic experiments.

Definition

An **experiment** or **trial** is any procedure that can be infinitely repeated and has a well-defined set of possible outcomes.
An **outcome** (denoted by ω) is a particular result of an experiment.
A **sample space** (denoted by Ω) is the set of all possible outcomes of an experiment (i.e. $\omega \in \Omega$).
An **event** is a subset of the sample space (e.g. $A \subset \Omega$), representing a specific outcome or a collection of outcomes.

Definition

Probability (denoted by $\mathbb{P}$) is a measure of the likelihood of an event occurring. It assigns a number between 0 and 1 to an event, where 0 indicates extreme unlikelihood, and 1 indicates certainty that the event will occur. In particular,

$$0 \leq \mathbb{P}(A) \leq \mathbb{P}(\Omega) = 1.$$

Example: Tossing a Coin

Example

When we throw a coin once, the possible outcomes are H, T (standing for 'head' and 'tail'). Therefore, $\Omega = \{H, T\}$.
There are 4 events one can consider: $\{H\}, \{T\}, \{H, T\}, \emptyset$.

Example: Rolling a Die

Example

Consider rolling a fair six-sided die. The possible outcomes are $1, 2, 3, 4, 5, 6$, so $\Omega = \{1, 2, 3, 4, 5, 6\}$.
Consider the event A of getting an even number: $A = \{2, 4, 6\}$.
Consider the event B of getting a prime number: $B = \{2, 3, 5\}$.
Then $\mathbb{P}(A) = \mathbb{P}(B) = \frac{1}{2}$ (think why).

Impossible Events

Remark

The event $\emptyset$ ('empty set') describes an **impossible event** (e.g. we throw a coin or a die with no outcome). Then

$$\mathbb{P}(\emptyset) = 0.$$

The Discrete Case

Definition

We start our course with the **discrete case**, when Ω is a finite set. To calculate the probability $\mathbb{P}(A)$ of an event $A \subset \Omega$, we use the following formula:

$$\mathbb{P}(A) = \frac{\text{number of outcomes that make } A}{\text{number of all outcomes}} = \frac{\sharp(A)}{\sharp(\Omega)}.$$

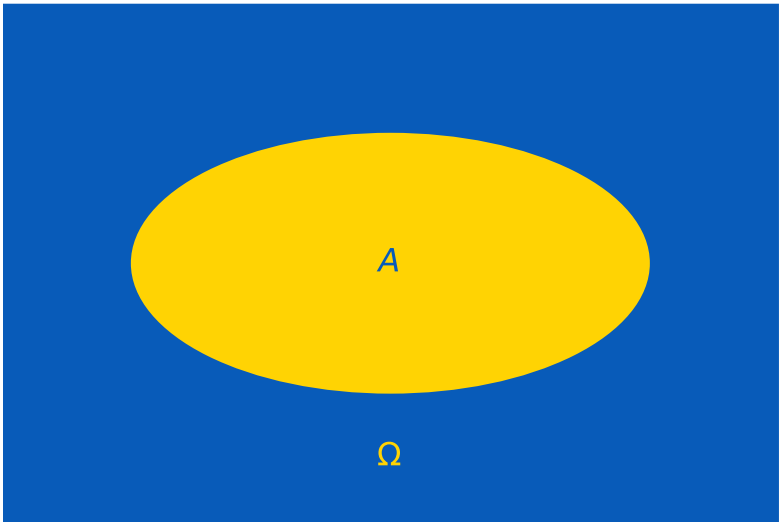


Figure 1: Visual representation of $\mathbb{P}(A) = \dfrac{\textbf{oval}}{\textbf{rectangle}}$

Example: Rolling a Die Twice

Example

Consider rolling a fair six-sided die twice. Then outcomes are $\omega = (a, b)$ where $a, b \in \{1, 2, 3, 4, 5, 6\}$, i.e.

$$\Omega = \{(a, b) \mid a, b \in \{1, 2, 3, 4, 5, 6\}\}.$$

Then $\sharp(\Omega) = 6 \cdot 6 = 36$. Let A be the event of having the sum of the numbers in the two rollings bigger than 10.
Then

$$A = \{(5, 6), (6, 5), (6, 6)\}.$$

Therefore,

$$\mathbb{P}(A) = \frac{3}{36} = \frac{1}{12}.$$

Example: Drawing a Card

Example

Consider drawing a card from a standard deck of 52 playing cards. The sample space Ω is the set of all 52 pairs of the form vS, where $v \in \{A, 2, 3, 4, \ldots, 10, J, Q, K\}$ is the value of a card, and $S \in \{\clubsuit, \diamondsuit, \spadesuit, \heartsuit\}$ is the card suit.
Let B be the event of drawing a red face card. Then

$$B = \{J\diamondsuit, Q\diamondsuit, K\diamondsuit, J\heartsuit, Q\heartsuit, K\heartsuit\},$$

and hence,

$$\mathbb{P}(B) = \frac{6}{52} = \frac{3}{26}.$$

Rules of Probability

We now turn to the algebra of events: combining, comparing, and conditioning on events.

The Sum and Product of Events

Definition

The **sum** of events A and B is the event $A + B$ (also denoted $A \cup B$ or $A \vee B$) which occurs iff *either* A occurs *or* B occurs *or they both* occur.

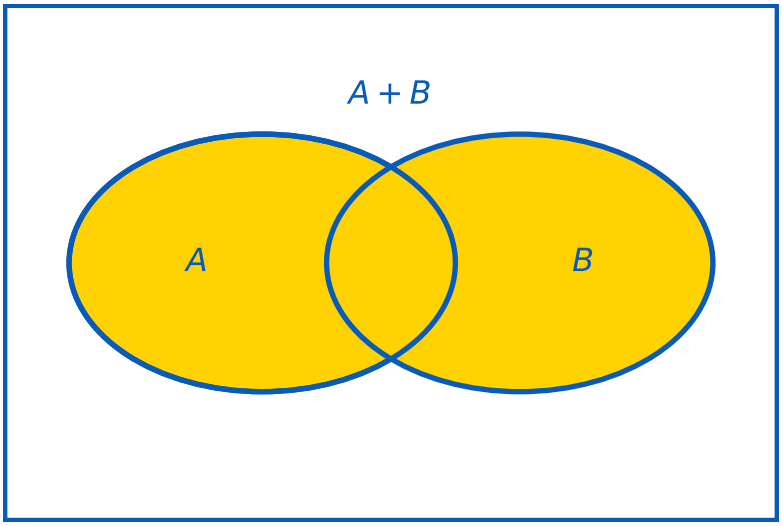


Figure 2: Visual representation of $A + B$

Definition

The **product** of events A and B is the event AB (also denoted $A \cap B$ or $A \wedge B$) which occurs iff *both* A *and* B occur.

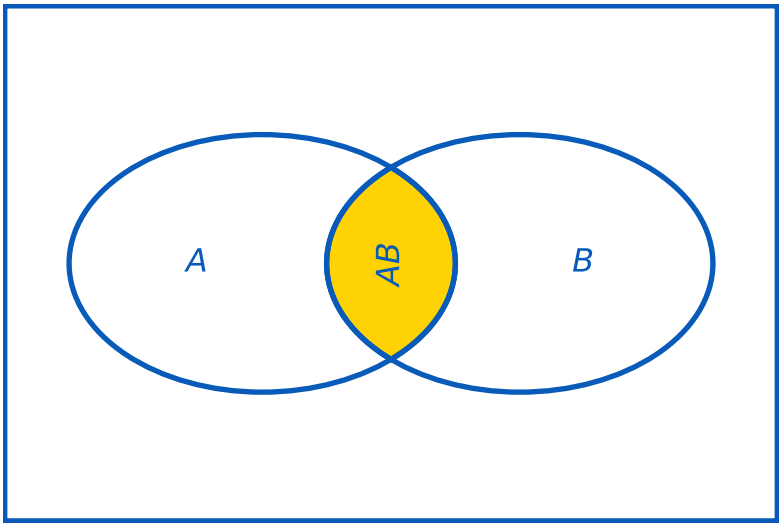


Figure 3: Visual representation of AB

Comparing $A + B$ and AB with A, B

Remark

$A + B$ occurs under *more* outcomes than either of A or B alone:

$$\mathbb{P}(A + B) \geq \mathbb{P}(A), \quad \mathbb{P}(A + B) \geq \mathbb{P}(B).$$

AB occurs under *less* outcomes than each of A or B alone:

$$\mathbb{P}(AB) \leq \mathbb{P}(A), \quad \mathbb{P}(AB) \leq \mathbb{P}(B).$$

The Addition Rule

Definition

The **addition rule** states that

$$\mathbb{P}(A + B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(AB).$$

It can be easily interpreted using the visual representations of $\mathbb{P}(A + B)$ and $\mathbb{P}(AB)$.

Example for the addition rule

Example

There is a standard deck of 52 playing cards. Find the probability of drawing either a red card or a face card (king, queen, or jack) from the deck in a single draw.
Solution: Let A be the event of drawing a red card, and B be the event of drawing a face card. Overall, there are 26 red cards, 12 face cards, and 6 red face cards. Therefore,

$$\mathbb{P}(A+B) = \frac{26}{52} + \frac{12}{52} - \frac{6}{52} = \frac{32}{52} = \frac{8}{13}.$$

Mutually Exclusive Events

Remark

Events A and B are called **mutually exclusive events** if only one of them may happen, i.e. if $AB = \emptyset$. In this case $\mathbb{P}(AB) = 0$, and the addition rule takes the form

$$\mathbb{P}(A + B) = \mathbb{P}(A) + \mathbb{P}(B).$$

The Complement of an Event

Definition

The **complement** to an event A is the event A^c which occurs iff A *does not* occur. Since $\mathbb{P}(\Omega) = 1$, one has

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

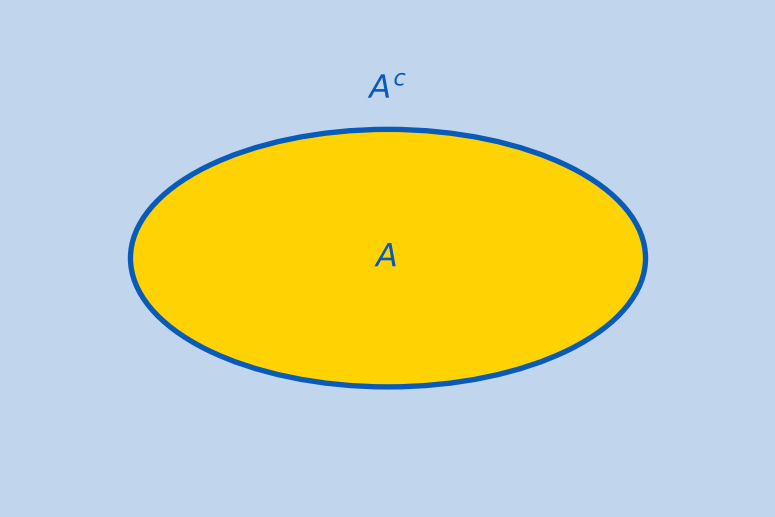


Figure 4: Visual representation of A^c

Example

A fair six-sided die is rolled three times. Find the probability that the total score (the sum of three trials) will be at least 4 (event A).
Solution: The sample space Ω consists of all triples (a, b, c) with $a, b, c \in \{1, 2, \dots, 6\}$. Thus, $\sharp(\Omega) = 6^3 = 216$. The total score is 4 or more in all cases but the case $(1, 1, 1)$. Therefore, the answer is:

$$\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = 1 - \frac{1}{216} = \frac{215}{216}.$$

Conditional Probability

Definition

Conditional probability $\mathbb{P}(A \mid B)$ is the probability of an event A occurring given that event B has already occurred, so we assume that $\mathbb{P}(B) \neq 0$. The formula is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(AB)}{\mathbb{P}(B)}.$$

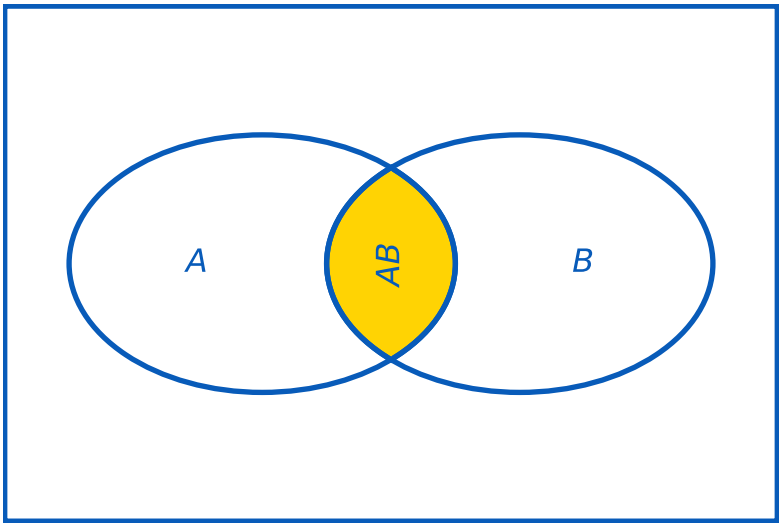


Figure 5: The event AB, which underlies conditional probability

The Multiplication Rule

Definition

The **multiplication rule** follows immediately from the formula for the conditional probability:

$$\mathbb{P}(AB) = \mathbb{P}(B)\,\mathbb{P}(A \mid B) = \mathbb{P}(A)\,\mathbb{P}(B \mid A).$$

Remark

The multiplication rule can be generalised for the product of several events, e.g.

$$\mathbb{P}(ABC) = \mathbb{P}(A)\,\mathbb{P}(B \mid A)\,\mathbb{P}(C \mid AB).$$

Example: Drawing Marbles from a Bag

Example

In a bag of 20 marbles, 8 are red, and 12 are green. Three marbles are drawn from the bag without replacement. What is the probability that they all are of the same colour?
Solution: we need the probability that either $A = (r, r, r)$ or $B = (g, g, g)$ holds. Note that A and B are *mutually exclusive events*. By the multiplication rule,

$$\mathbb{P}(A) = \frac{8}{20} \cdot \frac{7}{19} \cdot \frac{6}{18},$$

and

$$\mathbb{P}(B) = \frac{12}{20} \cdot \frac{11}{19} \cdot \frac{10}{18}.$$

Therefore, by the addition rule (for mutually exclusive events),

$$\begin{aligned} \mathbb{P}(A + B) &= \mathbb{P}(A) + \mathbb{P}(B) \\ &= \frac{8 \cdot 7 \cdot 6}{20 \cdot 19 \cdot 18} + \frac{12 \cdot 11 \cdot 10}{20 \cdot 19 \cdot 18} \\ &= \frac{1656}{6840} = \frac{23}{95}. \end{aligned}$$

Independence of Events

Definition

An event A is said to be **independent** of an event B if the occurrence of B does not affect the probability of occurrence of A. In other words, A is independent of B iff

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Example

A fair coin is tossed twice. Let A: a head appeared in the first toss, and B: a tail appeared in the second toss. Then A and B are independent: $A = \{HH, HT\}$, $B = \{HT, TT\}$, and hence $AB = \{HT\}$, giving $\mathbb{P}(AB) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(A)\mathbb{P}(B)$.

Remark

If A is independent of B then B is independent of A, and

$$\mathbb{P}(AB) = \mathbb{P}(A)\,\mathbb{P}(B).$$

Remark

If three (or more) events are pairwise independent — A and B independent, B and C independent, A and C independent — it still may be that they are not independent in total, and then, in general, $\mathbb{P}(ABC) \neq \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$.

The Law of Total Probability

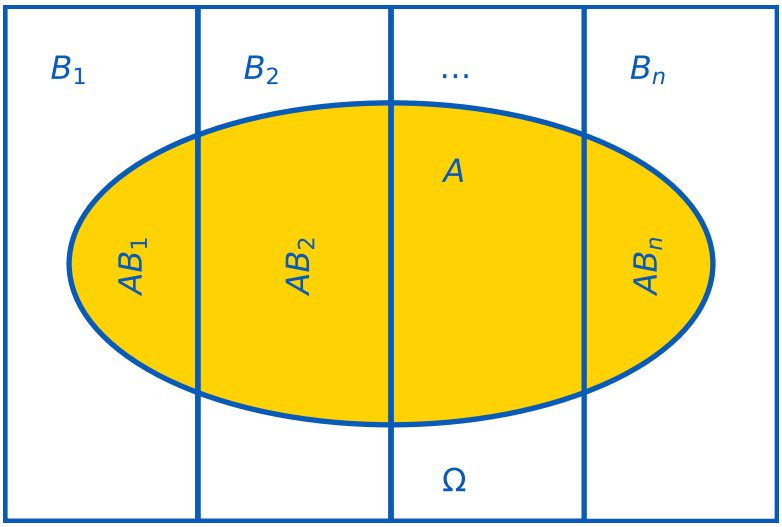


Figure 6: Note that $A = AB_1 + ... + AB_n$

Definition

Let $B_1, \dots, B_n$ be pairwise exclusive events (i.e. $B_i B_j = \emptyset$ for all $i \neq j$) such that $B_1 + B_2 + ... + B_n = \Omega$ with $\mathbb{P}(B_i) \neq 0$ (it is said then that $B_1, \dots, B_n$ form a *partition* of Ω). Then the **law of total probability** holds:

$$\begin{aligned}\mathbb{P}(A) = \mathbb{P}(A \mid B_1)\,\mathbb{P}(B_1) \\ + ... + \mathbb{P}(A \mid B_n)\,\mathbb{P}(B_n).\end{aligned}$$

Remark

There is also a modification of the law of total probability for conditional probabilities. If $B_1, \dots, B_n$ are as above and $\mathbb{P}(C) \neq 0$, then

$$\begin{aligned}\mathbb{P}(A \mid C) = \mathbb{P}(A \mid B_1 C)\mathbb{P}(B_1 \mid C) \\ + ... + \mathbb{P}(A \mid B_n C)\mathbb{P}(B_n \mid C).\end{aligned}$$

Bayes' Rule

From the conditional-probability definition,

$$\begin{aligned}\mathbb{P}(B)\mathbb{P}(A \mid B) &= \mathbb{P}(AB) = \mathbb{P}(BA) \\ &= \mathbb{P}(A)\mathbb{P}(B \mid A).\end{aligned}$$

This implies the following important statement.

Definition

Let $\mathbb{P}(A) \neq 0$ and $\mathbb{P}(B) \neq 0$. Then **Bayes' rule** (a.k.a. Bayes' formula or Bayes' theorem) holds:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A)\mathbb{P}(B \mid A)}{\mathbb{P}(B)}.$$

It describes the *a posteriori* probability of the event A, after an experiment with the known outcome B, using the *a priori* information about the outcome B.

Remark

If $A_1, \ldots, A_n$ form a *partition* of Ω (i.e. $A_1 + \ldots + A_n = \Omega$ and $A_i A_j = \emptyset$ for $i \neq j$) with $\mathbb{P}(A_j) \neq 0$, then we can rewrite Bayes' rule as follows, for $\mathbb{P}(B) \neq 0$:

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(A_i)\mathbb{P}(B \mid A_i)}{\sum_{j=1}^{n} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)}.$$

Example: A Disease Test (Bayes' Rule)

Example

A patient has taken a test for a rare disease. The prevalence of the disease in the population is known to be very low, only 0.1%. The test correctly identifies the disease in 95% of cases when it's present. The test incorrectly indicates the presence of the disease in 3% of cases where it's not actually present. The patient has just received a positive test result for the disease. What is the probability that he actually has the disease?

Example

Solution: Let D denote the event of having the disease, $\mathbb{P}(D) = 0.001$ (0.1%). Let T denote the event of a positive test result. Then

$$\mathbb{P}(T|D) = 0.95, \qquad \mathbb{P}(T|D^c) = 0.03.$$

By definition, D and D^c form a partition of Ω. Then

$$\begin{aligned}\mathbb{P}(D|T) &= \frac{\mathbb{P}(T|D) \cdot \mathbb{P}(D)}{\mathbb{P}(T|D) \cdot \mathbb{P}(D) + \mathbb{P}(T|D^c) \cdot \mathbb{P}(D^c)} \\ &= \frac{0.95 \cdot 0.001}{0.95 \cdot 0.001 + 0.03 \cdot 0.999} \approx 0.0306.\end{aligned}$$

So, given a positive test result, the probability that the patient actually has the disease is just 3.06%.

Sensitivity to the False-Positive Rate

Remark

The result is quite small because the false-positive rate (3%) is quite high relative to the disease prevalence. Fixing prevalence at 0.1% and sensitivity at 95%, consider $\mathbb{P}(D \mid T)$ as a function $f(p)$ of the false-positive rate $p \in [0, 1]$:

$$f(p) = \frac{0.95 \cdot 0.001}{0.95 \cdot 0.001 + p \cdot 0.999}.$$

Clearly $f(0) = 1$ and $f(1) \approx 0.00095$. As p increases, $f(p)$ decreases — so to get a high probability that the patient actually has the disease given a positive result, the test needs a very small false-positive rate.

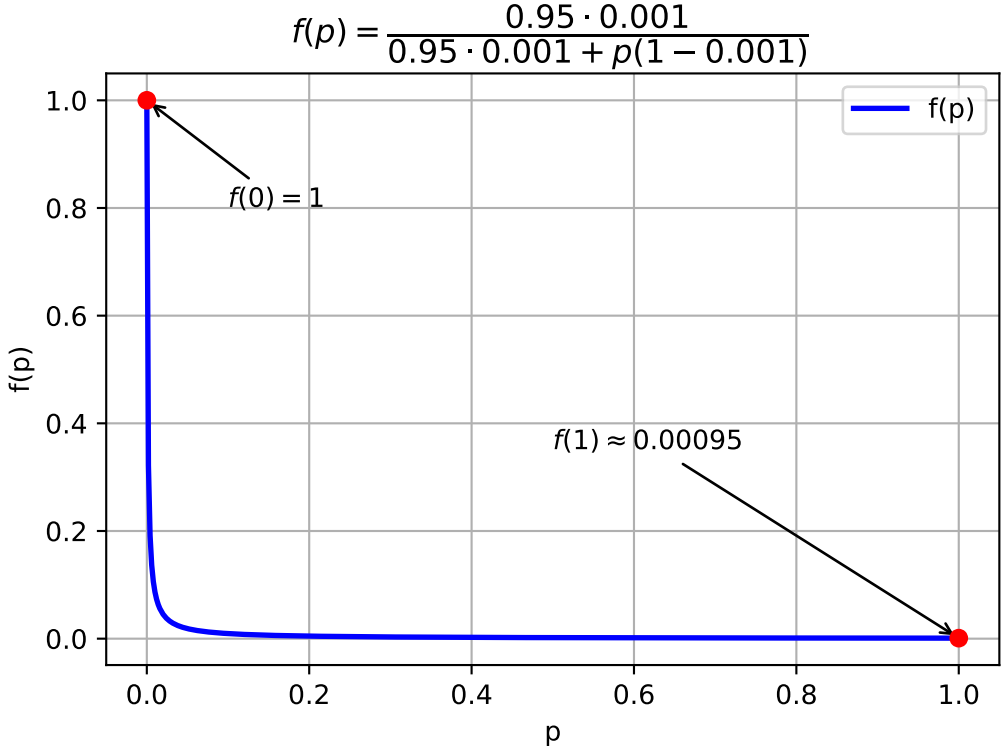


Figure 7: $f(p)$ against the false-positive rate p